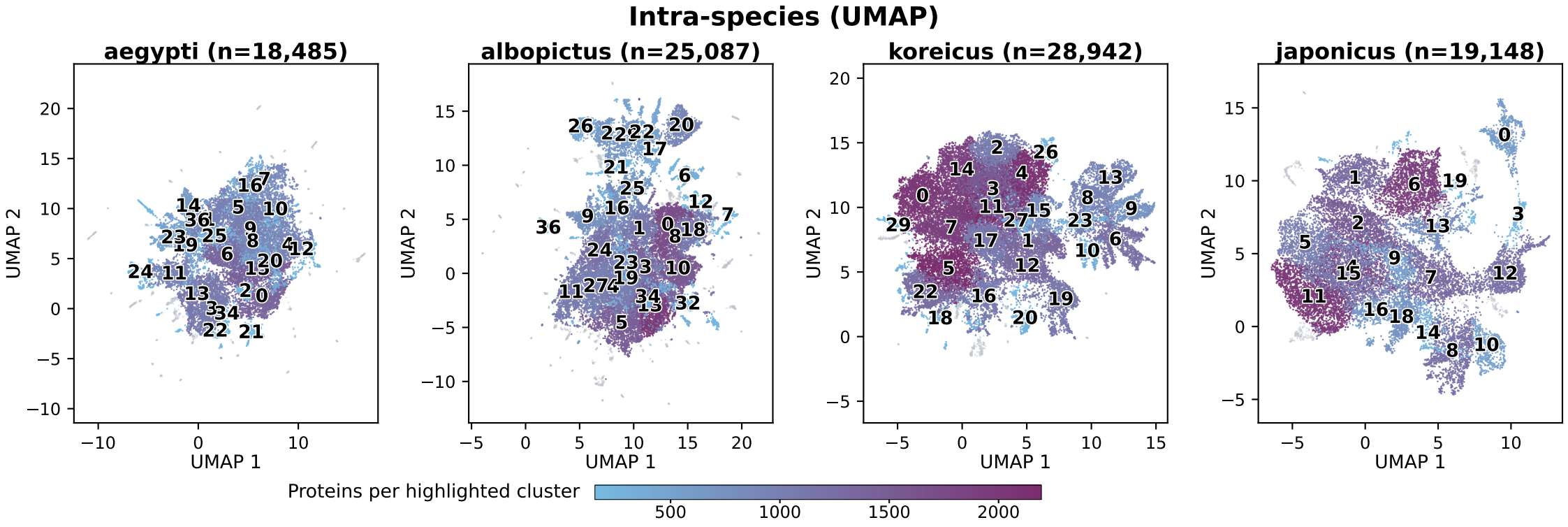
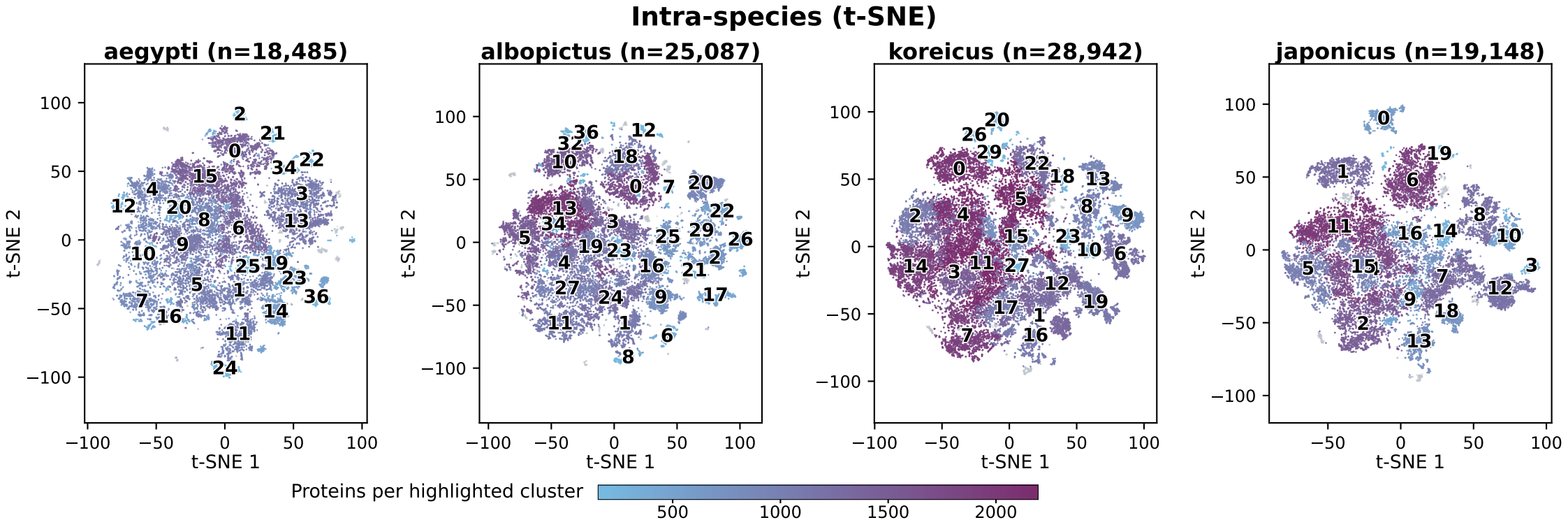
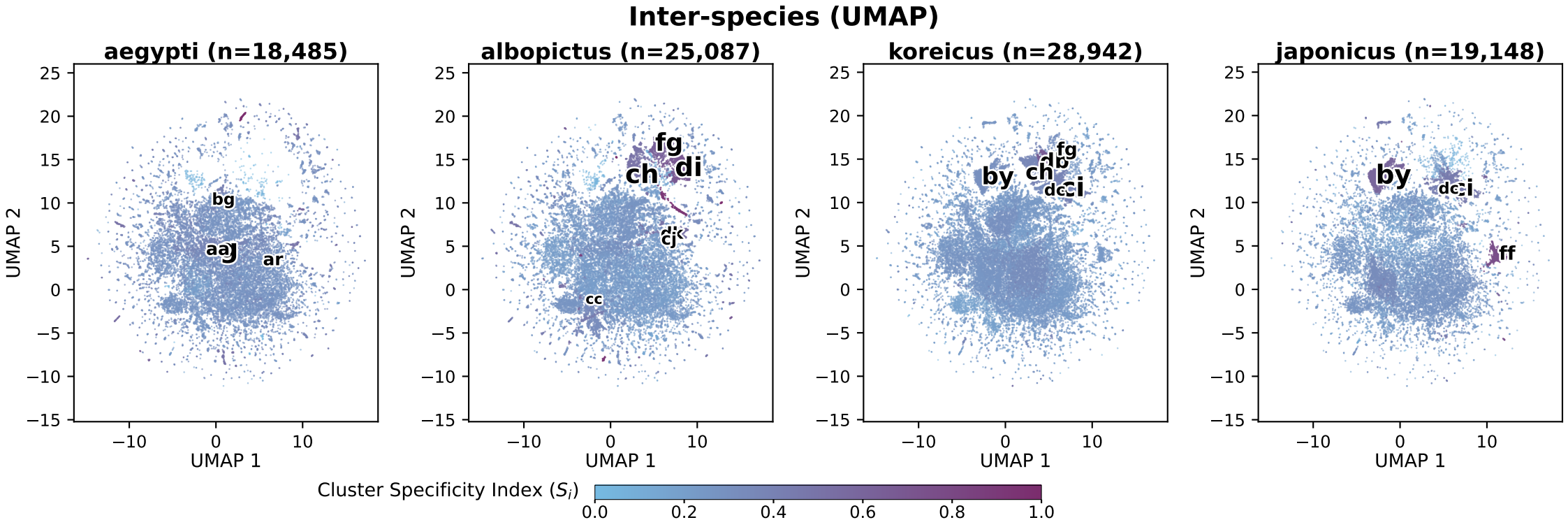
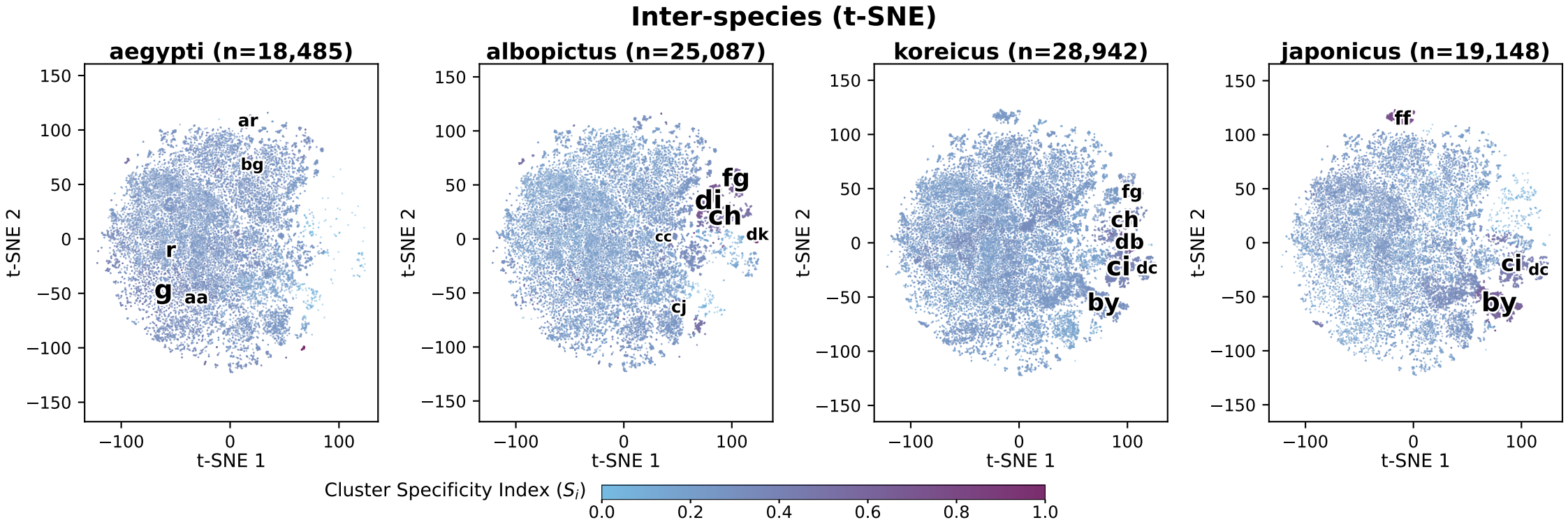
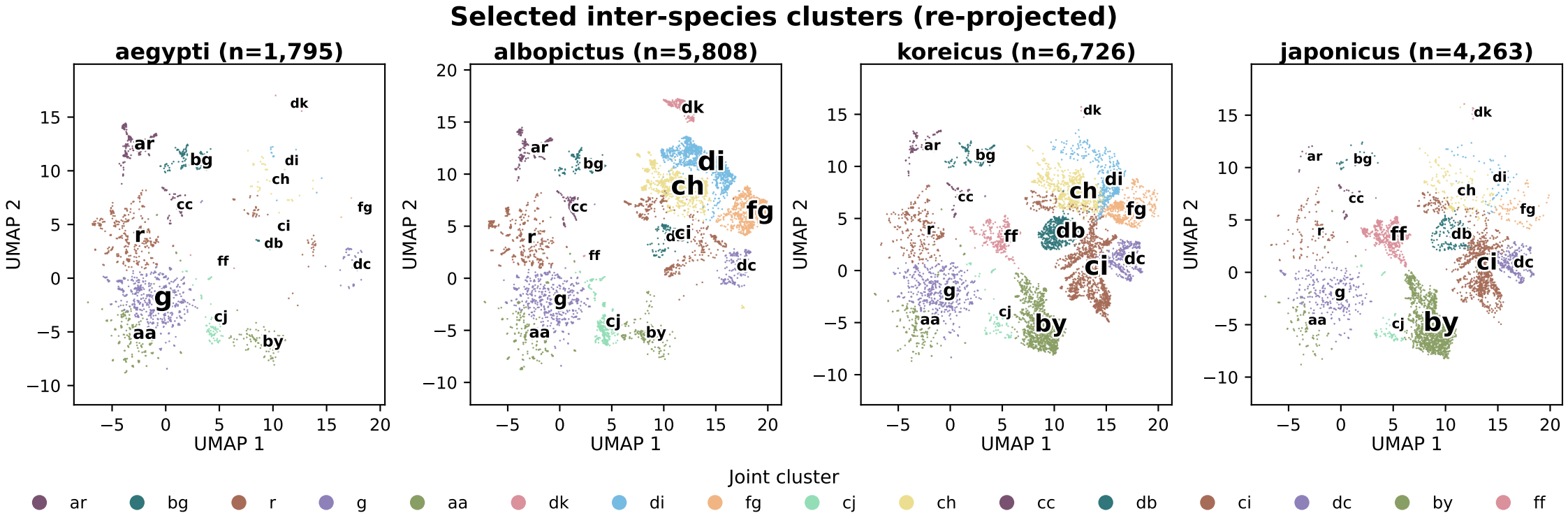

Intra-species (UMAP)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
UMAP 1
UMAP 2
Proteins per highlighted cluster
500
1000
1500
2000


Intra-species (t-SNE)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
t-SNE 1
t-SNE 2
Proteins per highlighted cluster
500
1000
1500
2000


Inter-species (UMAP)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
UMAP 1
UMAP 2
bg
aaj
ar
fg
di
ch
cj
cc
by
ff
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Inter-species (t-SNE)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
t-SNE 1
t-SNE 2
ar
bg
r
g
aa
fg
di
ch
dk
cc
cj
fg
ch
db
ci
dc
by
ff
ci
dc
by
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Selected inter-species clusters (re-projected)
aegypti (n=1,795)
albopictus (n=5,808)
koreicus (n=6,726)
japonicus (n=4,263)
UMAP 1
UMAP 2
Joint cluster
ar
bg
r
g
aa
dk
di
fg
cj
ch
cc
db
ci
dc
by
ff

Intra- to Inter-species cluster fluxes

aegypti (n=18,485, k=66)

0
13
9
3
1
5
4
10
7
8
16
12
21
34

ar (182)
bg (134)
r (341)
g (555)
aa (197)

albopictus (n=25,087, k=71)

5
3
20
9
2
29
22
17
26
6
32

dk (265)
di (1,035)
fg (902)
cj (284)
ch (1,023)
cc (100)

koreicus (n=28,942, k=40)

11
4
1
12
22
6
19
8
13
9
10
23

db (724)
ci (1,260)
ch (846)
dc (331)
by (1,036)
fg (494)

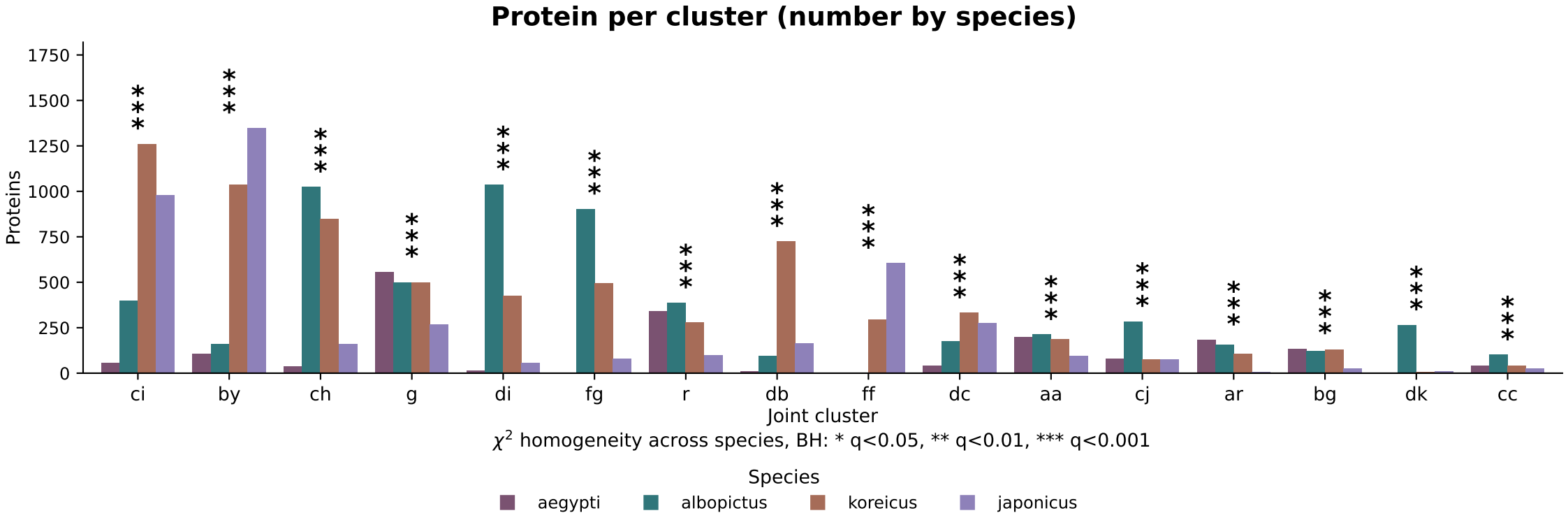
japonicus (n=19,148, k=29)

6
2
7
8
12
0
10
14
3

ff (604)
by (1,348)
ci (979)
dc (274)

Cluster Specificity Index (S_i); black border: $S_i \geq$ 0.80

0.0 0.2 0.4 0.6 0.8 1.0


Protein per cluster (number by species)
Proteins
1750
1500
1250
1000
750
500
250
0
ci
by
ch
g
di
fg
r
db
ff
dc
aa
cj
ar
bg
dk
cc
Joint cluster
χ^2 homogeneity across species, BH: * q<0.05, ** q<0.01, *** q<0.001
Species
aegypti
albopictus
koreicus
japonicus